Lecture 05 — Linear and affine functions

Wed Sep 2

Last updated: September 2, 2026.

Reading: Boyd 2.1 and 2.3 (linear and affine functions; regression model). Related objectives: Lectures. Schedule.

Learning objectives

By the end of class, students should be able to:

1. Distinguish linear from affine functions in examples.
2. Test superposition numerically on simple functions.
3. Construct a function $a^\mathsf{T} x + b$ for a contextual quantity.
4. Evaluate a model by hand and computationally.

Fill in

A function $f : \mathbb{R}^n \to \mathbb{R}$ is **linear** if for all n-vectors x, y and scalars α, β,

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y).$$

This property is called ______________.

Equivalently, linear functions satisfy **homogeneity** $f(\alpha x) = \alpha f(x)$ and **additivity** $f(x + y) = f(x) + f(y)$.

Every linear function can be written as an inner product: $f(x) =$ ____________ for some fixed n-vector a.

A function is **affine** if it has the form $f(x) = a^\mathsf{T} x + b$ for some n-vector a and scalar b. The number b is the ____________ (or intercept).

Affine functions satisfy a restricted superposition rule: $f(\alpha x + \beta y) = \alpha f(x) + \beta f(y)$ only when $\alpha + \beta =$ _______.

If $b \neq 0$, the function $f(x) = a^\mathsf{T} x + b$ is affine but ______________ linear.

The average $\text{avg}(x) = \frac{1}{n}(x_1 + \cdots + x_n)$ is ____________; the maximum entry $\max(x_1, \ldots, x_n)$ is not (for $n \geq 2$).

A regression model $\hat{y} = \beta^\mathsf{T} x + v$ predicts a scalar from features x. The entry β_i tells how $\hat{y}$ changes when feature x_i increases by one, holding other features fixed. The offset v is the predicted value when ________________.

Worked in class

W1. Classify each function below as **linear**, **affine but not linear**, or **neither**. If linear, write $f(x) = a^\mathsf{T} x$.

(a) $f(x) = x_n - x_1$ (Boyd 2.1(b)).

(b) $f(x) = 3x_1 - 2x_2 + 7$.

(c) $f(x) = \max(x_1, x_2)$ for $n = 2$.

W2. Superposition check. For $f(x) = \max(x_1, x_2)$, let $x = (1, -1)$, $y = (-1, 1)$, $\alpha = \beta = \frac{1}{2}$. Compute $f(\alpha x + \beta y)$ and $\alpha f(x) + \beta f(y)$. What does this show?

W3. A 10-vector r stores course grades: $r_1, \dots, r_8$ are homework scores (0–10 each), r_9 is midterm (0–120), r_10 is final (0–160). Total score on 0–100 is 25% homework, 35% midterm, 40% final. Write $s = w^\mathsf{T} r$ and give w (Boyd 1.10 style).

W4. Profit per unit is $p = (4, 2, -1)$ dollars for three items (item 3 is a loss leader). Monthly sales are $s = (100, 50, 30)$ units. Total profit is $p^\mathsf{T} s$ (Boyd 1.8). Compute it and interpret item 3's sign.

Check yourself

C1. Boyd 2.1(a): $f(x) = \max_k x_k - \min_k x_k$ (spread). Linear or not? One sentence why.

C2. Boyd 2.1(e): $f(x) = x_n + (x_n - x_{n-1})$ (extrapolation). Linear or not? If linear, find a with $f(x) = a^\mathsf{T} x$.

C3. Is $g(x) = 2x_1 - x_3 + 5$ linear, affine, or neither? Write it as $a^\mathsf{T} x + b$ if affine.

C4. Lachniet Ch. 1 Ex. 2(b): $u = (2, -4, 0)$, $v = (3, 1.5, -7)$. Find $d = u^\mathsf{T} v$ (dot product). What does $d = 0$ suggest about direction?

C5. A café bill is $f(x) = p^\mathsf{T} x$ with $p = (3.50, 2.00, 1.25)$ and $x = (2, 1, 0)$ (drinks ordered). Compute $f(x)$ and state whether f is linear or affine.

Answers (Check yourself)

C1. Not linear (nor affine): spread fails superposition (Boyd uses $x = (1, -1)$, $y = (-1, 1)$, $\alpha = \beta = 1/2$).

C2. Linear: $f(x) = x_n + (x_n - x_{n-1}) = 2x_n - x_{n-1}$, so $a = (0, \dots, 0, -1, 2)$ with $a_{n-1} = -1$, $a_n = 2$.

C3. Affine but not linear: $a = (2, 0, -1)^\mathsf{T}$, $b = 5$.

C4. $d = 2 \cdot 3 + (-4) \cdot 1.5 + 0 = 6 - 6 = 0$. The vectors are orthogonal (perpendicular in 3D).

C5. $f(x) = 3.50 \cdot 2 + 2.00 \cdot 1 + 1.25 \cdot 0 = 9.00$ dollars. Linear (also affine with $b = 0$).